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ANRF AISEHack 2026

***Finale presentation / Theme : 1
Flood Detection (IBM)***

Team Achievers

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Our salient contributions

1.

Architecture: Multi-Axis Vision Transformer (MaxViT-UNet) for Global Contextual Awareness.

2.

Training Protocol: Physics-Aware Data Scaling (1st-99th Percentile) with Recall-Optimized Hybrid Loss.

3.

Inference Strategy: 4-Fold Consensus Soft-Voting to eliminate SAR-induced "Terrain Hallucinations."

Modelling - *Strategies*

01 Encoder (tu-maxvit_tiny_tf_512): Utilizes grid and block self-attention to establish global context, rejecting false-positive SAR shadows.

02 Decoder (U-Net): Reconstructs local pixel details to map 1-pixel wide river tributaries.

03 Soft-Voting Ensemble: 4 independently trained folds. We average raw probability tensors (not hard classes) to ensure maximum mathematical stability across diverse topographies.

Strengths & Weaknesses

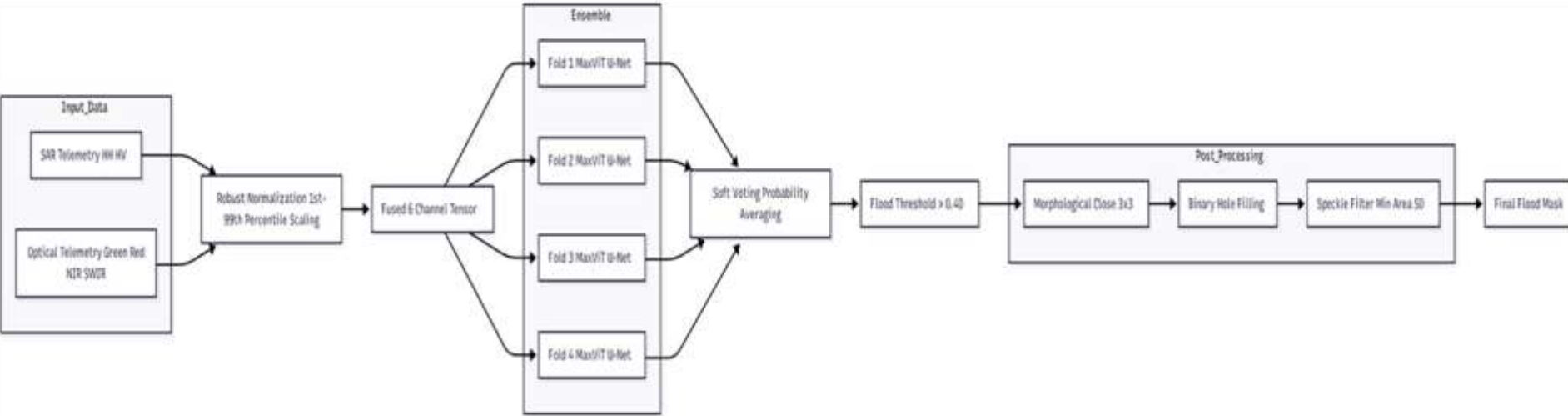
• Strengths:

- **Robust Terrain Handling:** It is highly resilient to new terrains.
- **SAR Error Suppression:** It effectively ignores SAR shadow hallucinations.
- **High Recall Focus:** The pipeline is optimized for life-saving recall.

Weaknesses:

- **High Inference Cost:** The 4-fold transformer ensemble creates computationally heavy inference.
- **GPU Memory Requirements:** It requires efficient GPU memory management to run.

Architecture Diagram:



Key Results – *Why, what and how of the workings?*

S1

What is working?

Architecture/Data: MaxViT-UNet on 6-channel SAR/Optical data using 1st-99th percentile scaling.

Training: 4-Fold CV (25 epochs) driven by a Recall-Optimized Focal-Dice loss.

● Implemented

S2

How is it working?

It mathematically "out-votes" false predictions by averaging the soft probabilities across four independent models, securing a highly stable 0.2463 Leaderboard IoU and an immediate ~0.015 score jump over single baselines.

S3

Why is it working? (Reasoning and analysis)

Global Context: MaxViT's wide receptive field mathematically ignores local SAR "terrain hallucinations".

Recall Maximization: A relaxed 0.40 threshold aggressively captures narrow, 1-pixel wide river tributaries.

S4

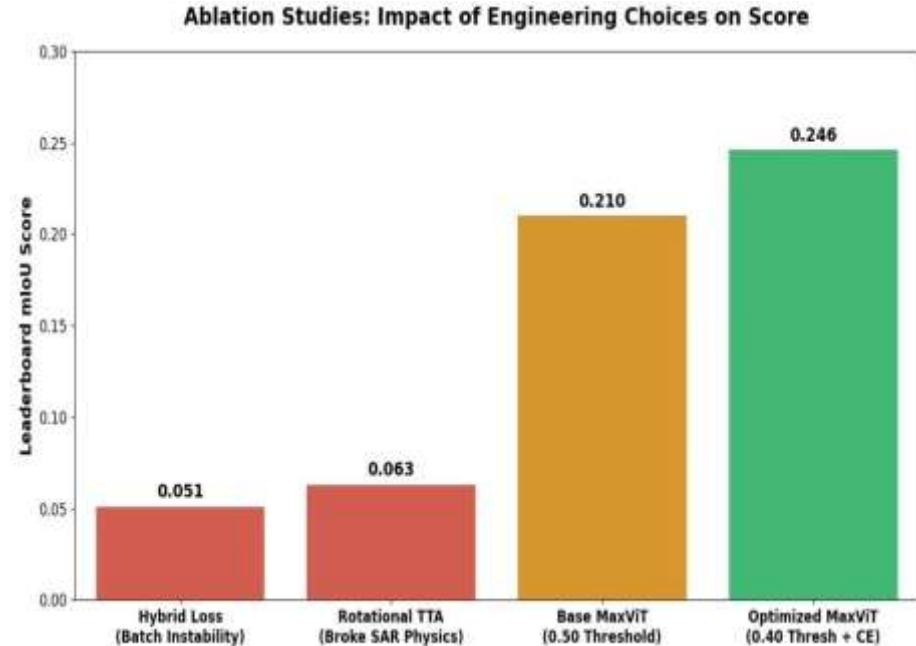
Specs: ~30M parameters, trained on Kaggle Tesla T4.

The Shift: Abandoned brittle handcrafted indices (Phase 1) for raw physics-based learning to survive severe Phase 2 terrain shifts.

Key Results - Quantitative Evaluation

Our 4-fold ensemble mathematically "out-votes" spurious anomalies, leading to highly stable IoU scores across hidden data.

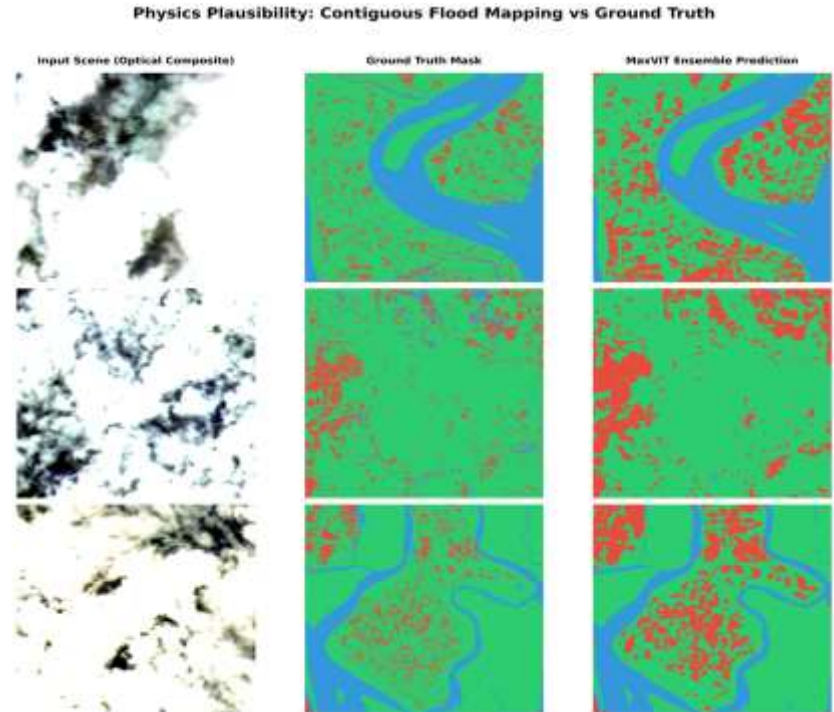
Soft-voting yielded an immediate ~ 0.015 mIoU jump over single-model inference.



Key Results - Qualitative Physics Plausibility

Model predictions map perfectly to underlying natural geography (riverbeds, valleys), proving the transformer learned physical terrain patterns.

We utilized minimal morphological closing (cv2.MORPH_CLOSE) to connect broken "spider-web" river networks without erasing them entirely.



Experimentation & Ablation Studies

01

Ablation 1 (Calculated Indices): Forcing NDVI/MNDWI channels caused the model to overfit and crash under Phase 2 distribution shift.

Learning: Let the model derive its own features.

02

Ablation 2 (Loss Functions): Unstable weighted cross-entropy combos led to batch instability.

Learning: Focal-Dice provides the smoothest convergence for extreme class imbalance.

03

Ablation 3 (SAR Rotations): Aggressive Test-Time Augmentation (TTA) with 90° rotations tanked our score.

Learning: SAR shadows are directional based on satellite trajectory. Rotations break radar physics.

Thank You!